API Testing Assignment

ReqRes API — Authentication, CRUD, and Automated Validation with Postman

Author: Grace UMWIZA

1. API Choice and Base URL

API used: ReqRes

Base URL: `https://reqres.in/api`

ReqRes was chosen because it provides CRUD-style endpoints with predictable, well-documented responses, making it well suited for demonstrating authentication, all HTTP verbs, environment variables, and automated testing in Postman. The /users endpoints are demo/testing fixtures, so requests return the expected HTTP status codes and response shapes rather than persisting changes to a real database.

2. Authentication Method Used

Method: API Key + Login Token

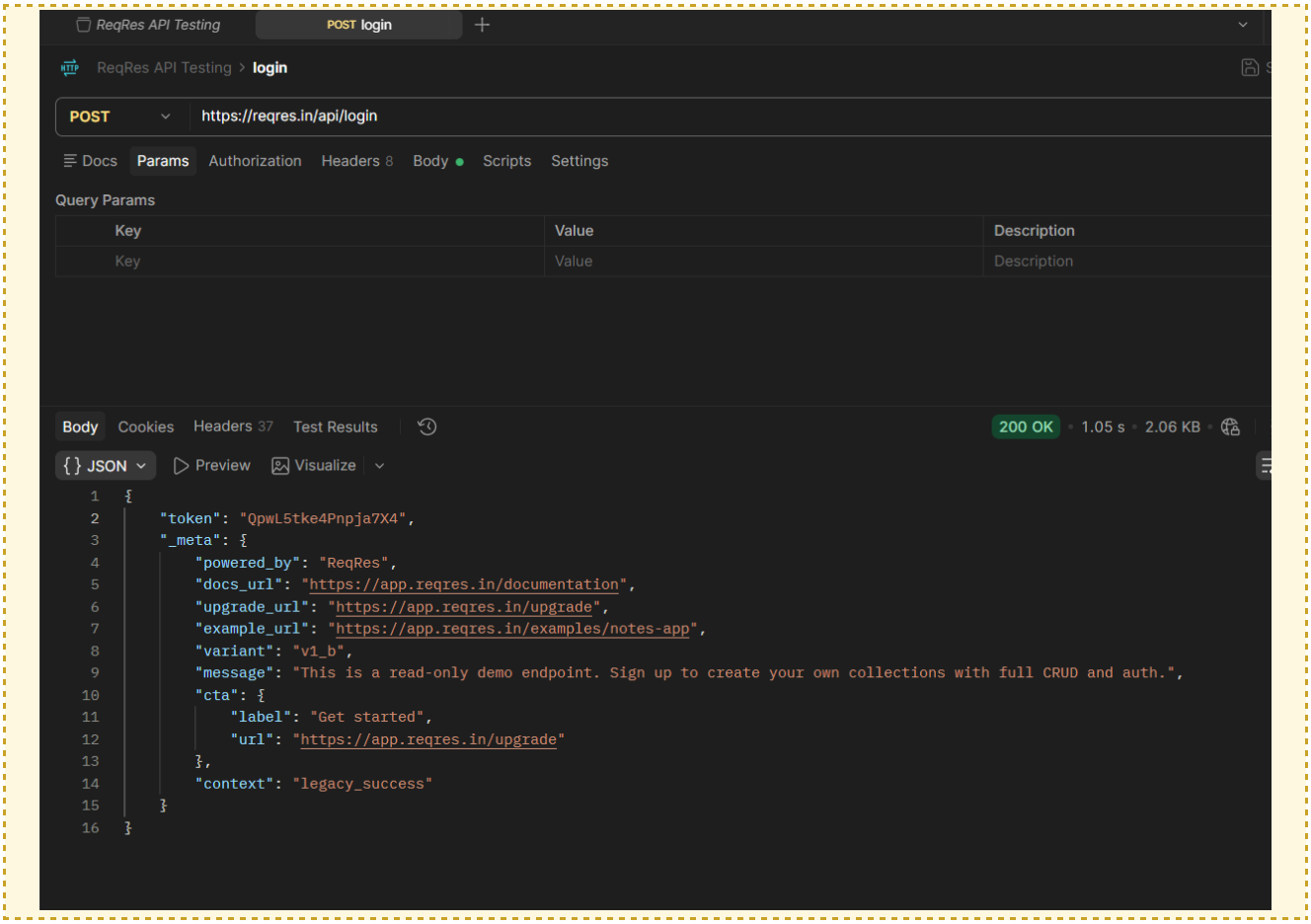
Two authentication mechanisms were demonstrated:

- **API key** — sent as a header on /users requests, stored in the Postman environment variable `api_key` and referenced as `{{api_key}}`. A Pre-request Script attaches it automatically:

```
pm.request.headers.upsert({
    key: "x-api-key",
    value: pm.environment.get("api_key")
});
```

- **Login token** — obtained via `POST {{base_url}}/login`. A successful response returns a bearer token, which is not hardcoded anywhere — it's captured automatically and stored in the environment variable `auth_token`.

Proof of authentication: `POST {{base_url}}/login` was sent with a valid email/password body. A 200 OK response containing a token confirms authentication is working correctly.



3. Endpoints Tested

#	Request	Method	Endpoint	Required Headers	Purpose
1	Login	POST	`/login`	`Content-Type: application/json`	Authenticate and retrieve a bearer token
2	Get All Users	GET	`/users?page=2`	`Authorization: Bearer {{auth_token}}` `x-api-key: {{api_key}}`	Retrieve a list of users
3	Get Single User	GET	`/users/{{user_id}}`	`Authorization: Bearer {{auth_token}}` `x-api-key: {{api_key}}`	Retrieve a specific user

4	Create User	POST	`/users`	`Authorization: Bearer {{auth_token}}`
				`x-api-key: {{api_key}}`
				`Content-Type: application/json`
					Create a new user

5	Update User	PUT	`/users/{{user_id}}`	`Authorization: Bearer {{auth_token}}`
				`x-api-key: {{api_key}}`
				`Content-Type: application/json`
					Fully update a user

#	Request	Method	Endpoint	Required Headers	Purpose

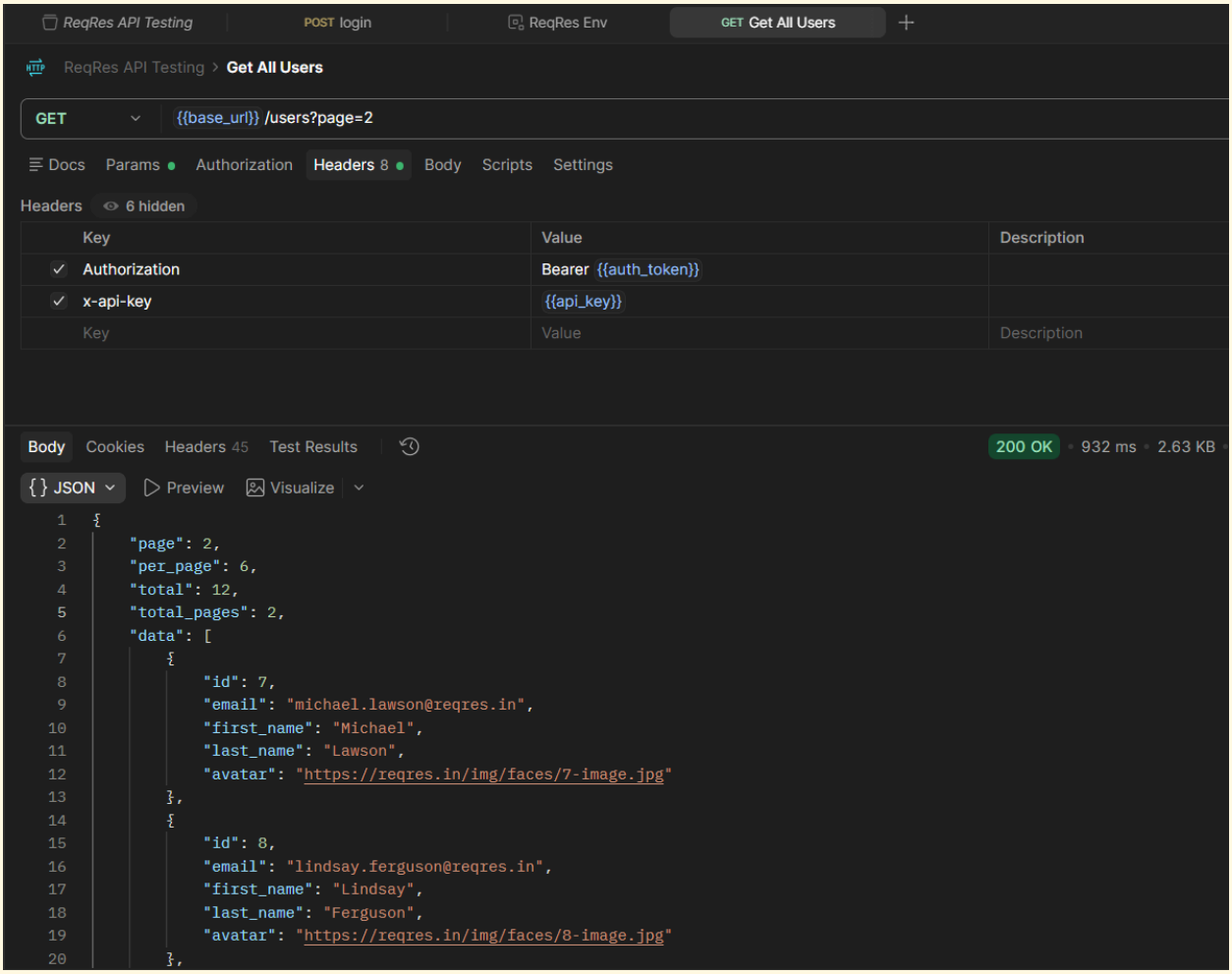
6	Patch Update User	PATCH	`/users/{{user_id}}`	`Authorization: Bearer {{auth_token}}`
				`x-api-key: {{api_key}}`
				`Content-Type: application/json`
					Partially update a user

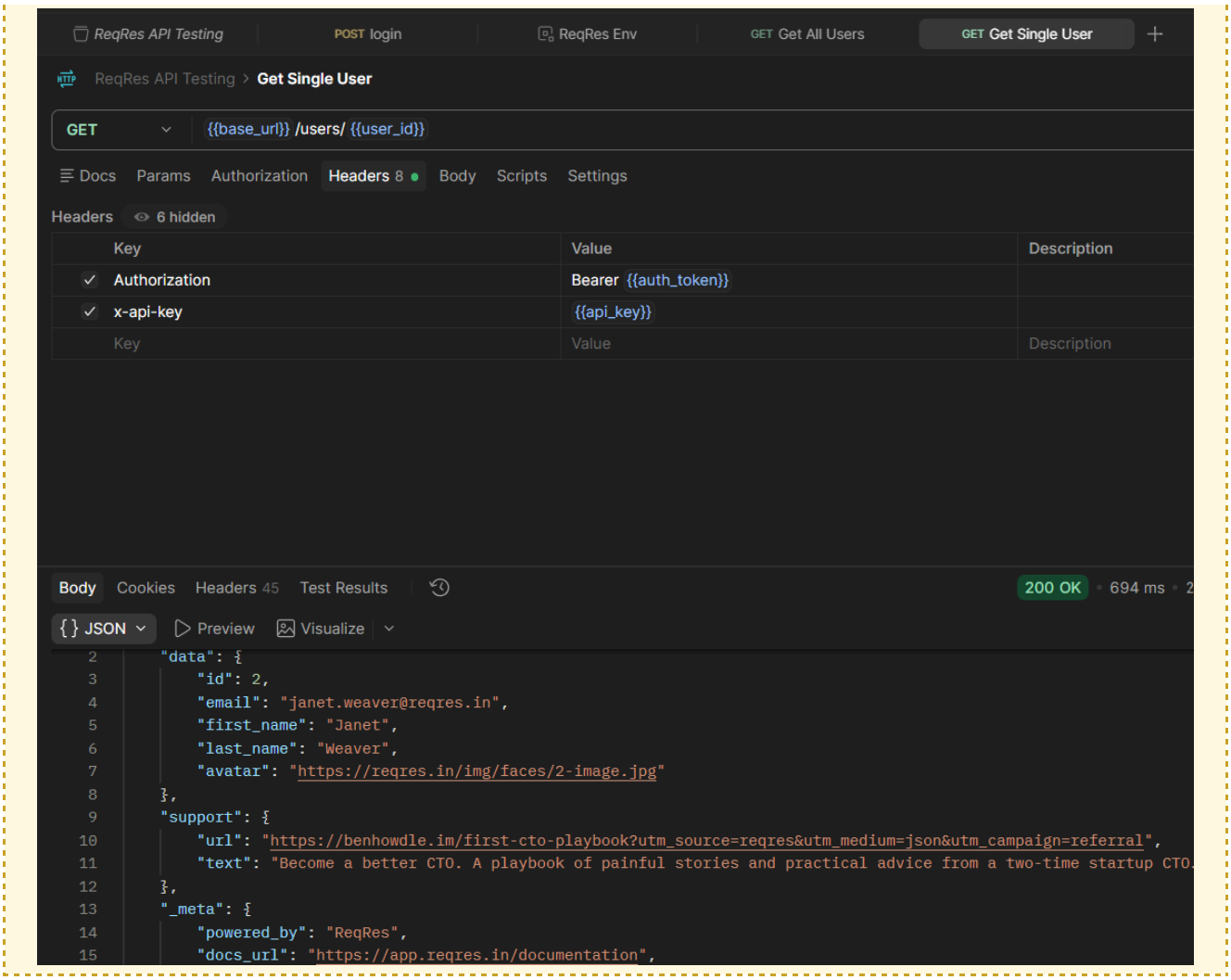
7	Delete User	DELETE	`/users/{{user_id}}`	`Authorization: Bearer {{auth_token}}` `x-api-key: {{api_key}}`	Delete a user

4. CRUD Operations — Explanation and Evidence

GET — Read

`GET {{base_url}}/users?page=2` retrieves a paginated list of users and returns 200 OK. `GET {{base_url}}/users/{{user_id}}` retrieves a single user by ID, also returning 200 OK.



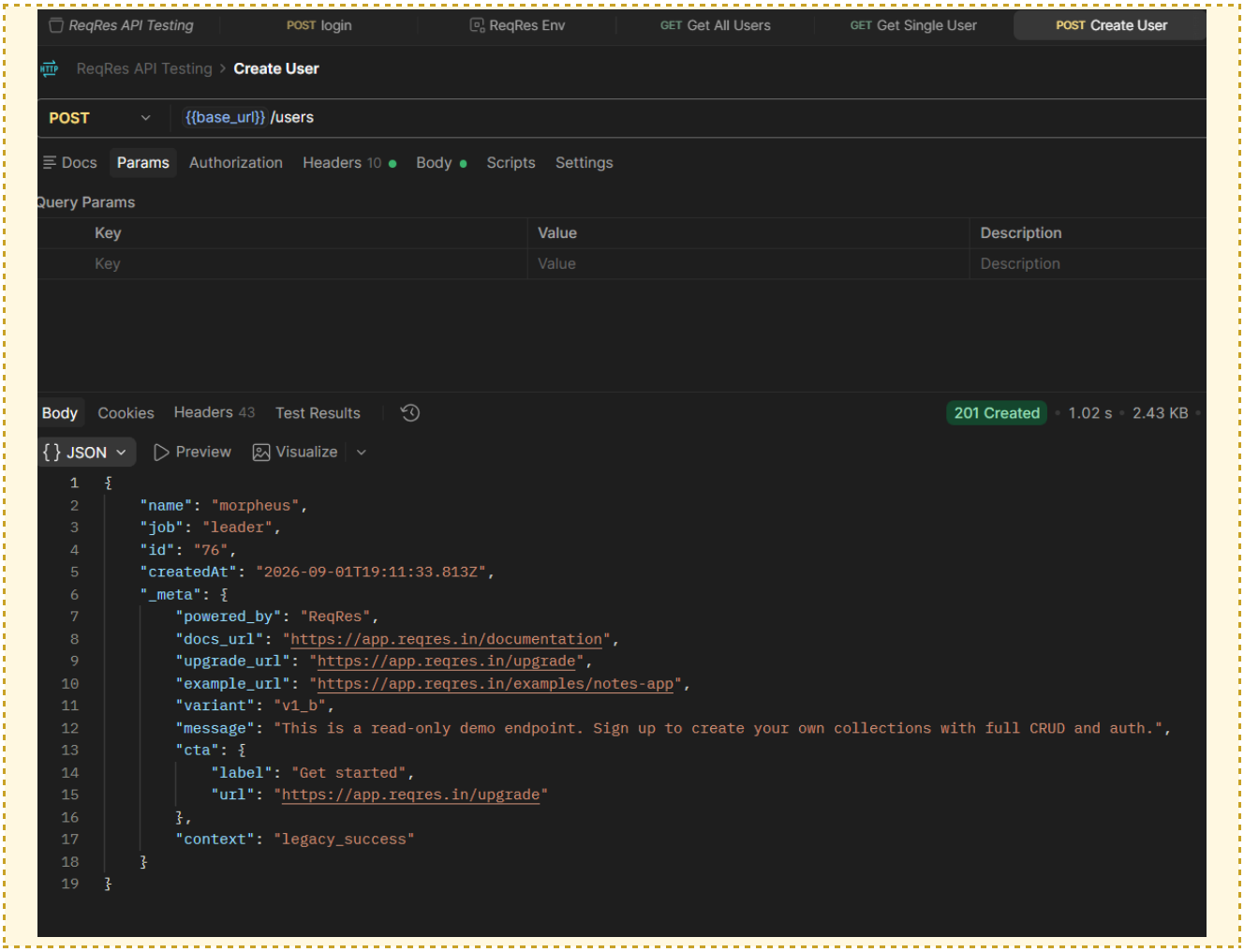


POST — Create

`POST {{base_url}}/users` creates a new user. Request body:

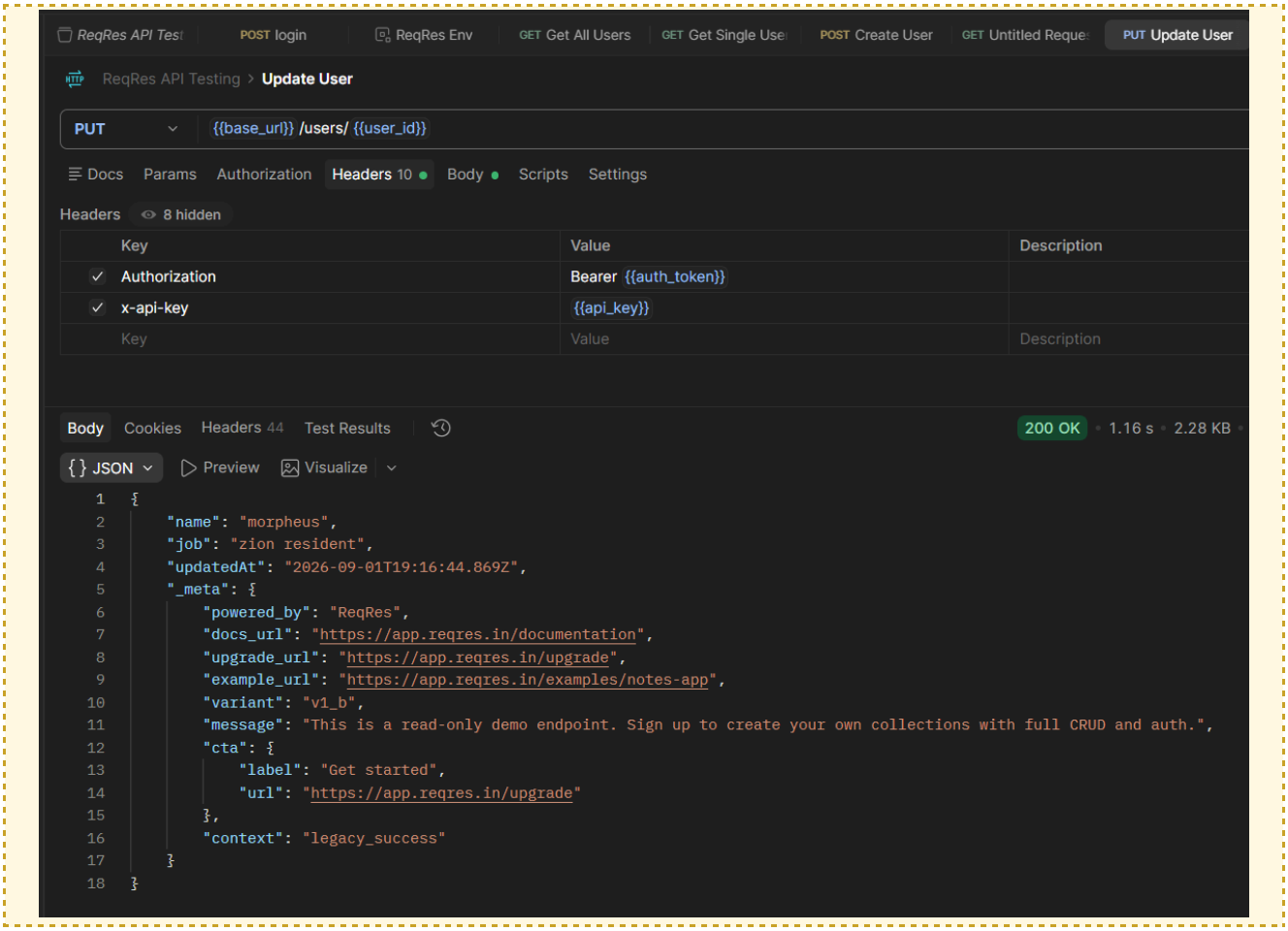
```
{
  "name": "morpheus",
  "job": "leader"
}
```

A successful request returns 201 Created along with the new user's id and a createdAt timestamp.



PUT — Full Update

`PUT {{base_url}}/users/{{user_id}}` replaces a user's full data. A successful update returns 200 OK with the updated fields and an updatedAt timestamp.

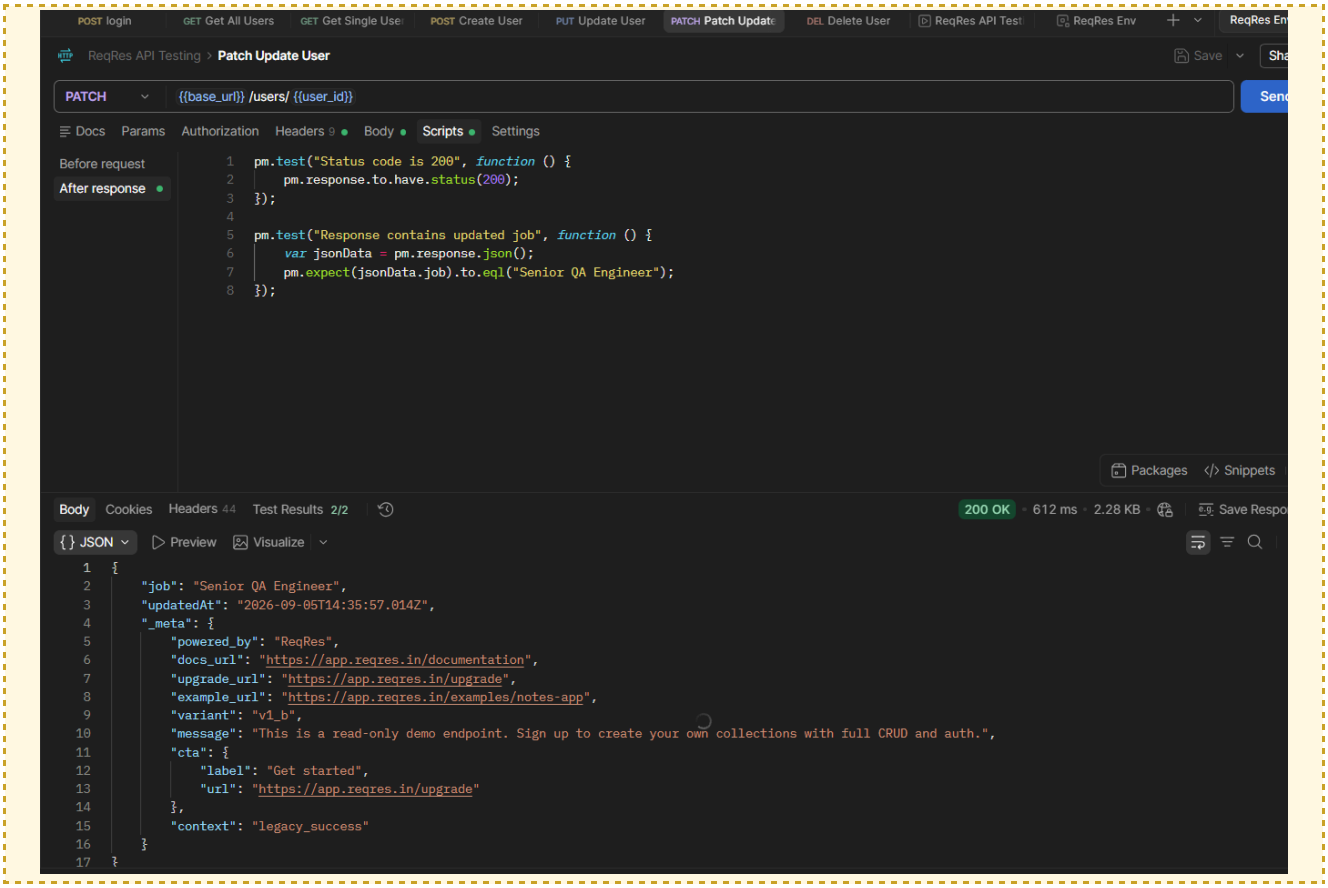


PATCH — Partial Update

`PATCH {{base_url}}/users/{{user_id}}` updates only the specified field(s). Request body:

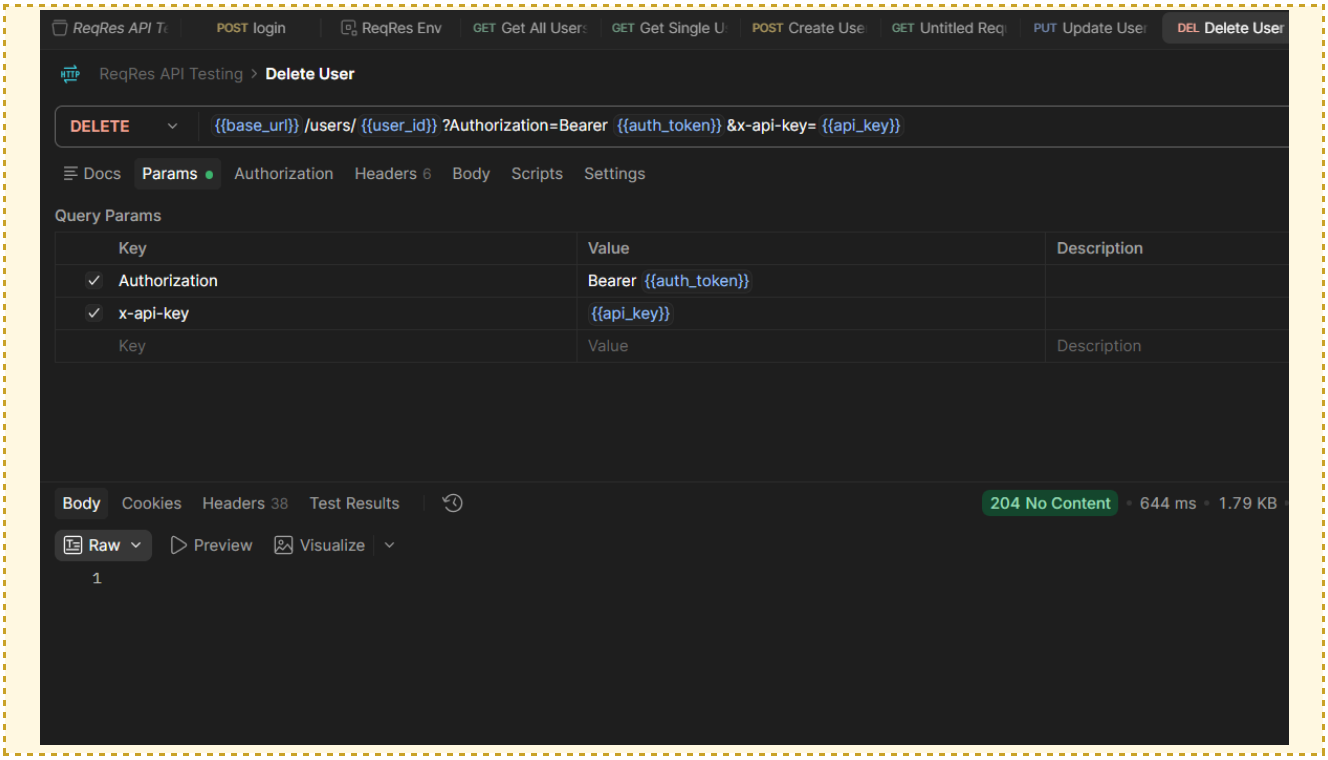
```
{
  "job": "Senior QA Engineer"
}
```

A successful update returns 200 OK, and the response confirms the updated value.



DELETE — Delete

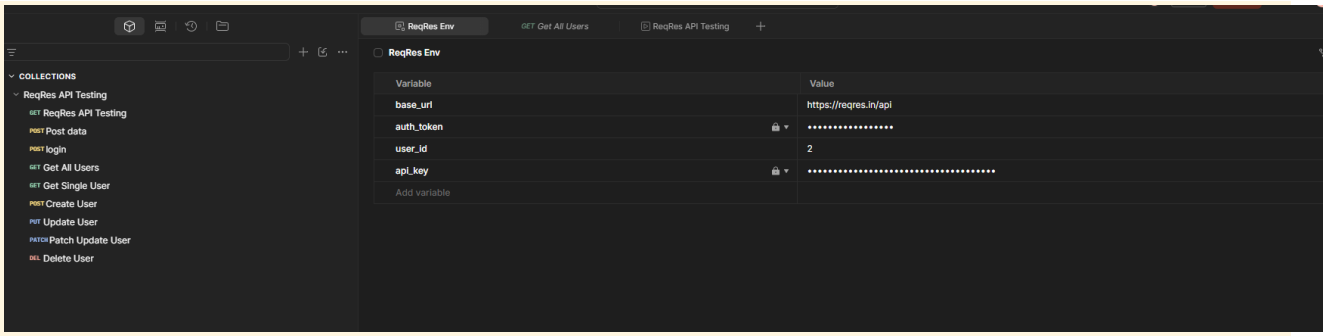
`DELETE {{base_url}}/users/{{user_id}}` removes the user. A successful deletion returns 204 No Content (an empty response body is expected and correct).



5. Environment Variables and Automation

A Postman environment named **ReqRes Env** was created to store values reused across requests instead of hardcoding them:

Variable	Purpose
base_url	https://reqres.in/api — used at the start of every request URL
auth_token	The bearer token returned from Login, used to demonstrate token-based auth
api_key	The API key used in the x-api-key header for /users requests
user_id	The ID of the user used by the Get Single User, Update, and Delete requests



Automation — saving the login token automatically

Rather than manually copying the token from the Login response, a test script was added to the Login request:

```
pm.test("Status code is 200", function () {
    pm.response.to.have.status(200);
});

pm.test("Response has token", function () {
    var jsonData = pm.response.json();
    pm.expect(jsonData.token).to.exist;
});

pm.environment.set("auth_token", pm.response.json().token);
```

This automatically extracts the token field from the response and saves it into `auth_token`, so any subsequent authenticated request can use `{{auth_token}}` with no manual steps in between.

Automation — Pre-request Script

A Pre-request Script was added to the Login request to automatically attach the current API key from the environment before the request is sent (shown in Section 2).

Per-request test scripts

Test scripts were added to each request to verify correct behavior, for example on the *Patch Update User* request:

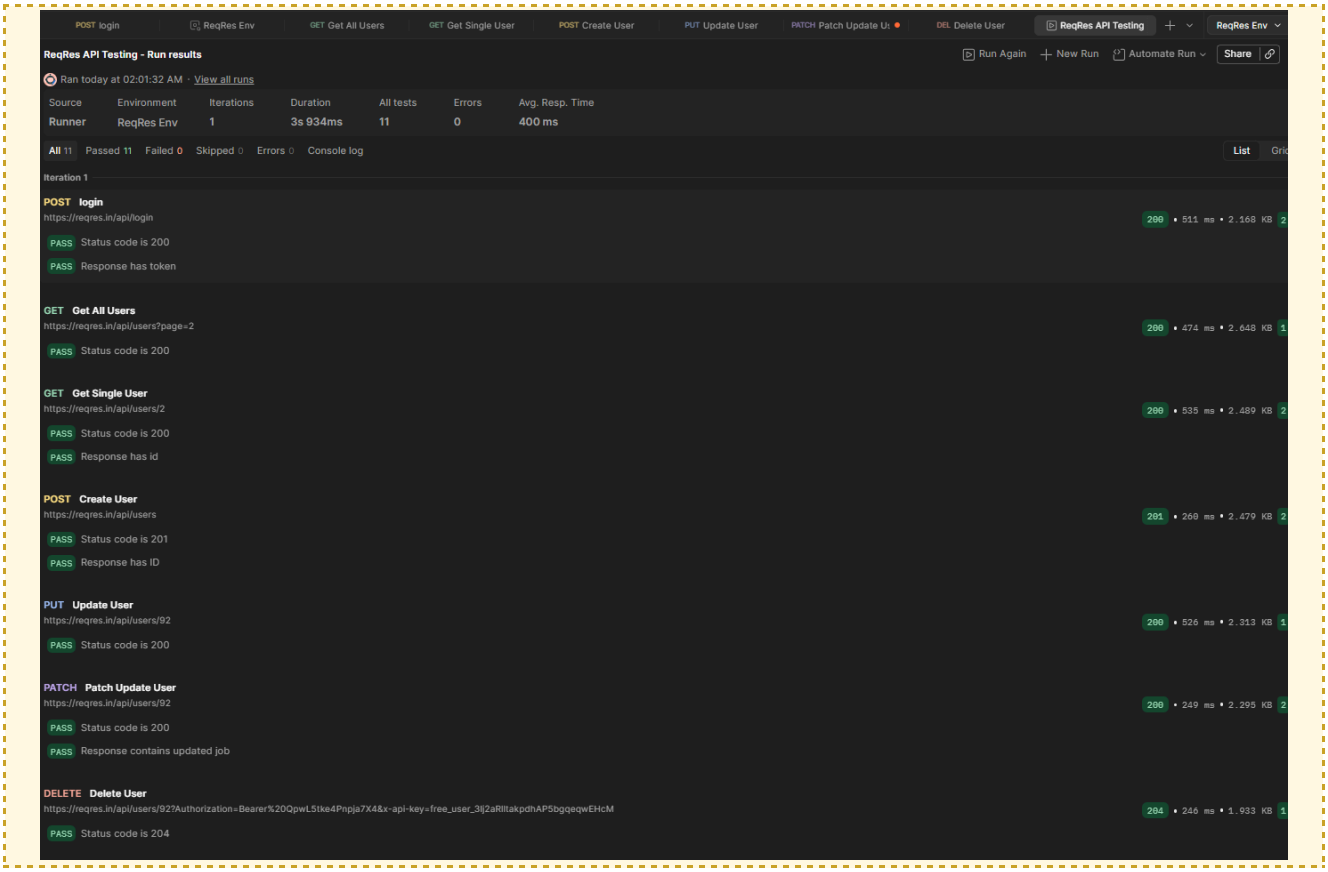
```
pm.test("Status code is 200", function () {
    pm.response.to.have.status(200);
});

pm.test("Response contains updated job", function () {
    var jsonData = pm.response.json();
    pm.expect(jsonData.job).to.eql("Senior QA Engineer");
});
```

6. Collection Runner

All seven requests (Login → Get All Users → Get Single User → Create User → Update User → Patch Update User → Delete User) were grouped into a single Postman Collection and executed in sequence using the Collection Runner, with the ReqRes Env environment selected. The run completed with 11/11 tests passed and 0 failures:

Request	Status	Test Result
Login	200	PASS
Get All Users	200	PASS
Get Single User	200	PASS
Create User	201	PASS
Update User	200	PASS
Patch Update User	200	PASS
Delete User	204	PASS

Collection Runner — 11/11 tests passed



7. Results Summary

Area	Result	Evidence
API exploration	Completed	Endpoint documentation and Postman requests
Authentication	Completed	Login, API key environment variable, token storage
CRUD	Completed	GET, POST, PUT, PATCH and DELETE
Environment variables	Completed	ReqRes Env with base_url, auth_token, api_key and user_id
Automation	Completed	Pre-request and post-response scripts
Testing	Completed	Postman response tests and Collection Runner

8. Conclusion

The ReqRes API was successfully explored and tested using Postman. The exercise demonstrated how API endpoints are documented and tested, how authentication information can be handled with environment variables, and how GET, POST, PUT, PATCH and DELETE requests can be validated. Postman scripts were also used to automate API-key setup and token storage, while response tests and the Collection Runner provided automated verification of the requests.

9. Deliverables in This Repository

```
.
├── README.md
├── collection/
│   └── reqres-api.postman_collection.json
├── environment/
│   └── reqres-env.postman_environment.json
└── docs/
    └── report.docx
```

Note: The exported environment file has all values (base_url, auth_token, api_key, user_id) left empty before being committed to this repository — no real credentials are included.

Note on screenshots: Full screenshots of every request/response (authentication, GET, POST, PUT, PATCH, DELETE, environment variables, and the Collection Runner run) are included directly in this report, at the point in each section where the corresponding request is discussed.